

Real-time Anomaly Detection in Space Communications using Large Language Models

SDDEC25-07

Meet the Team



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Problem Statement

We aim to close that gap by building a well-structured, flexible platform that can simulate realistic network behavior and use AI to detect and respond to problems in critical communication systems **achieve**.

Background

- CCSDS (Consultative Committee for Space Data Systems) & NASA Standards: published comprehensive standards for communication modeling
- LangChain & Hugging Face: very flexible as well as able to be run locally
- There is currently no existing system combining a network simulation with an LLM for space anomaly detection



Project Plan

The Project comprises 2 core modules:

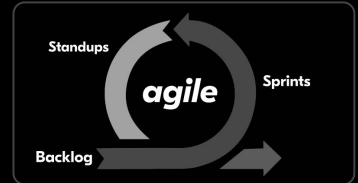
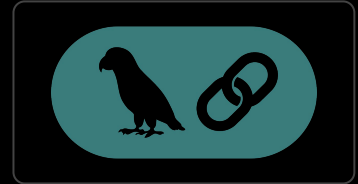
- ns-3-based satellite simulation environment
- AI evaluation pipeline (LangChain)

Together, these modules enable the realization of the complete system.

To support modularity and adaptability, an Agile development approach was adopted.

While the traditional Waterfall approach often lacks flexibility, Agile encourages:

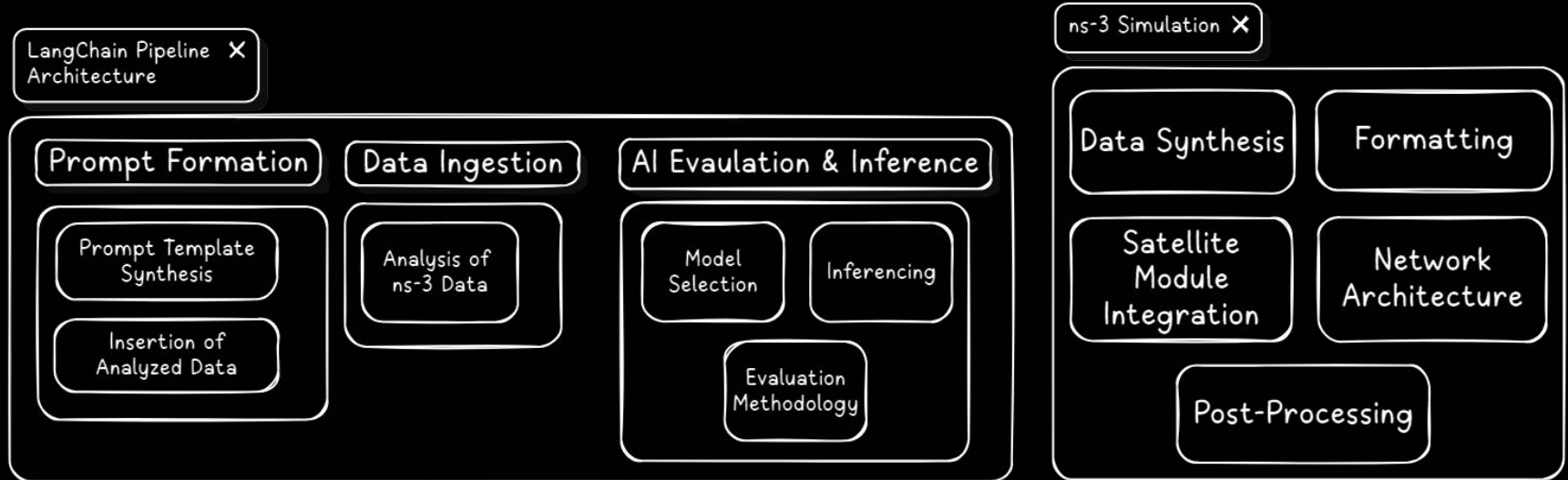
- Task deconstruction
- Adaptation to evolving requirements



Project Framework

The project is structured as a hierarchy of tasks/subtasks.

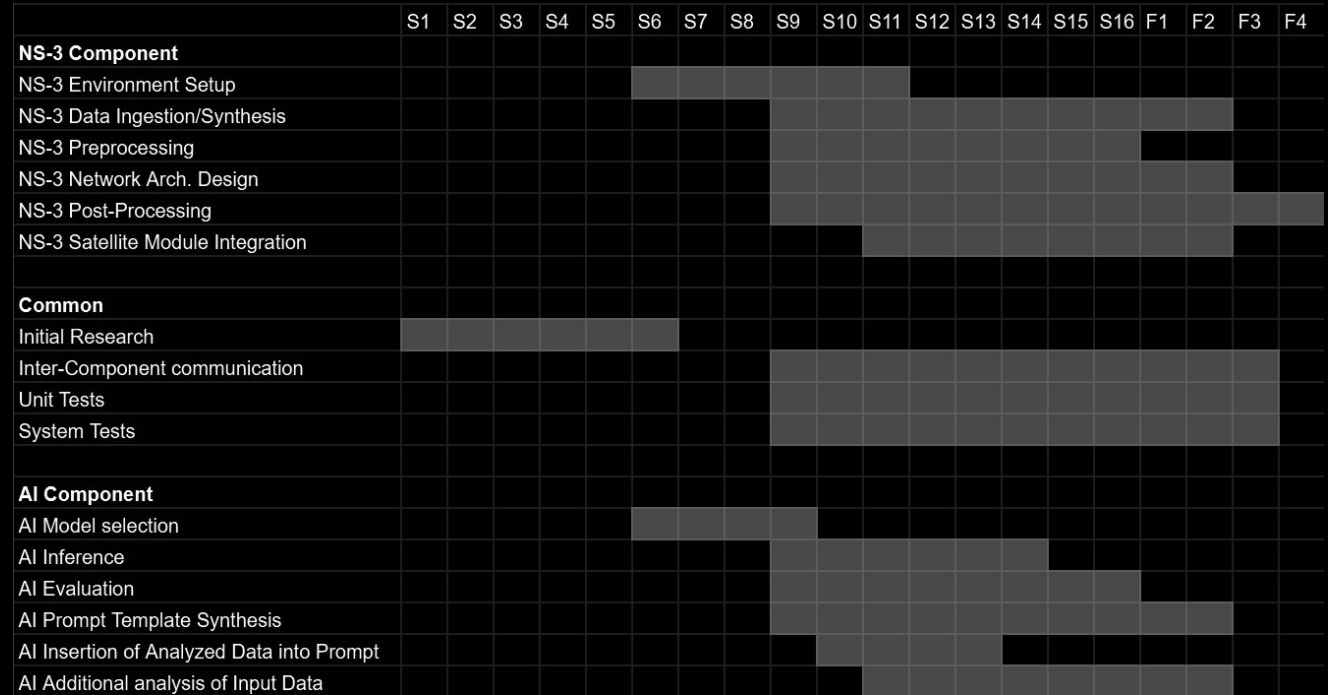
This enables parallel development when feasible to match the project's Agile nature.



Semester Gantt Chart

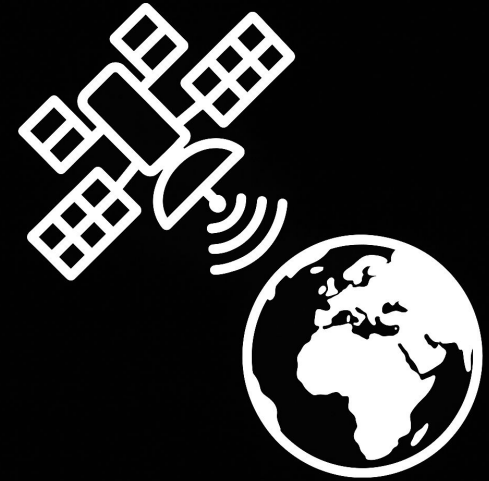
The following Gantt Chart was implemented to track project tasks and deliverables.

The Gantt chart enables structured progress and timely completion of milestones.



Project Abstract

The primary objective is to develop and integrate an LLM-based system into a Network Simulator 3 (ns-3) testbed, specifically configured for space communication scenarios. The system's performance is rigorously evaluated using metrics such as valid output formatting, false positive/negative rates, and response times.



Design Process

Simulation Architecture Selection

The team chose ns-3 as the core simulation framework.

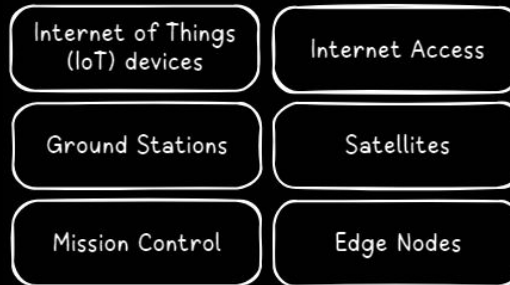
ns-3's network protocol support and modularity allows flexible simulation of real-world satellite communication scenarios and network behavior.



Node Types + Communication Path

The team designed a layered communication flow.

This structure reflects real satellite telemetry/control systems for accurate anomaly detection.



Data Input + Output Format

The team uses CSV + PCAP formats of the UNSW-NB15 dataset to evaluate the LLM anomaly detection pipeline.

This dataset provides structured examples of normal + malicious activity.

The system will later transition to ns-3-generated synthetic datasets, maintaining format consistency while shifting focus to domain-specific simulation data.

ns-3 — Design Process

Exploring multiple-node communication models is essential to evaluate trade-offs: realism — complexity — system scalability

Considering various architectures ensures: feasibility — robustness — alignment with space-based anomaly detection needs

The following designs were explored:

Direct-to-Internet IoT

Simplifies architecture but doesn't reflect real satellite routing paths.



Suitable for testing basic relay behavior but limited in scope for full system simulations.



More realistic, supports logging, edge analysis, and anomaly detection pipelines.

Mesh of Satellites for Handoff Simulation

Increases realism + complexity, simulating satellite-to-satellite relay before ground contact.

Hybrid Terrestrial and Space Network

Combines satellite uplinks with terrestrial edge nodes for distributed analysis.

The third option was selected as the base model. —> Balances realism + technical feasibility to directly support project's anomaly detection goals.

AI Pipeline — Design Process

Selecting an appropriate architectural design is a critical step in developing a functional and scalable AI system.

Each architecture presents trade-offs: feasibility — complexity— suitability

The following designs were evaluated:

ChatGPT Wrapper

High performance, cloud-based
Not suitable for real-time satellite
use due to API latency

Local AI using Hugging Face Transformers

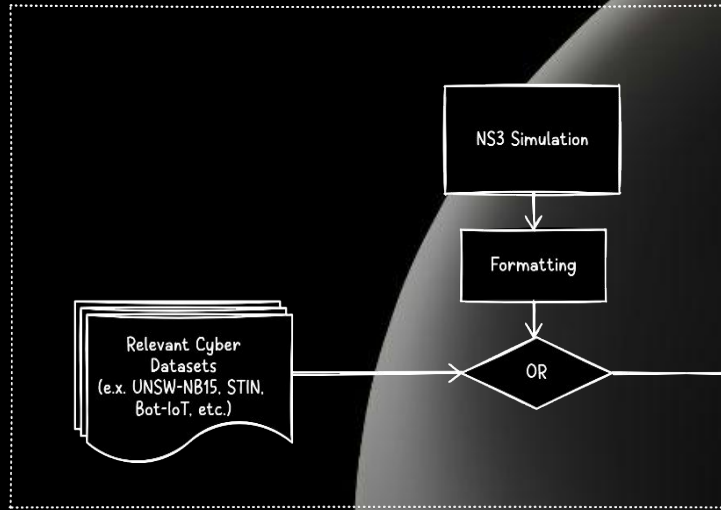
Lightweight, edge-compatible
Limited workflow control and
programmability

Local AI using LangChain with Hugging Face + DSPy Integration

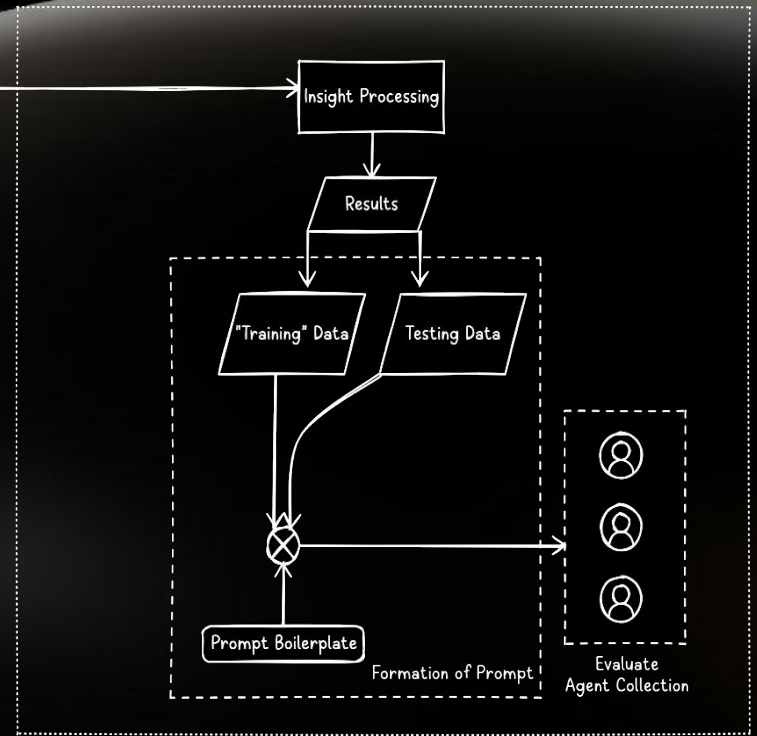
Modular pipeline design,
programmable logic, supports
structured reasoning
Higher implementation complexity

The third option was selected as the base model. —> Flexibility + control enables programmable anomaly detection within simulation system.

Project Architecture



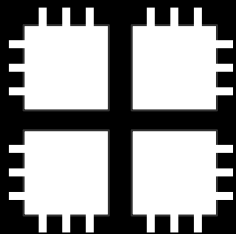
ns-3 Simulation



AI Pipeline

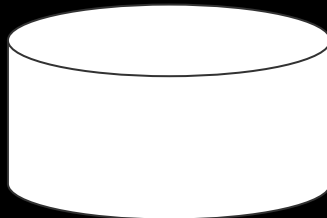
Technical Considerations

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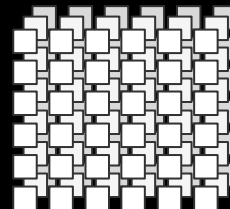
Multi-Core CPU
Minimize wait times

2



Large Storage
Cache local models

3



CUDA-Capable Card
Standard compute platform for AI



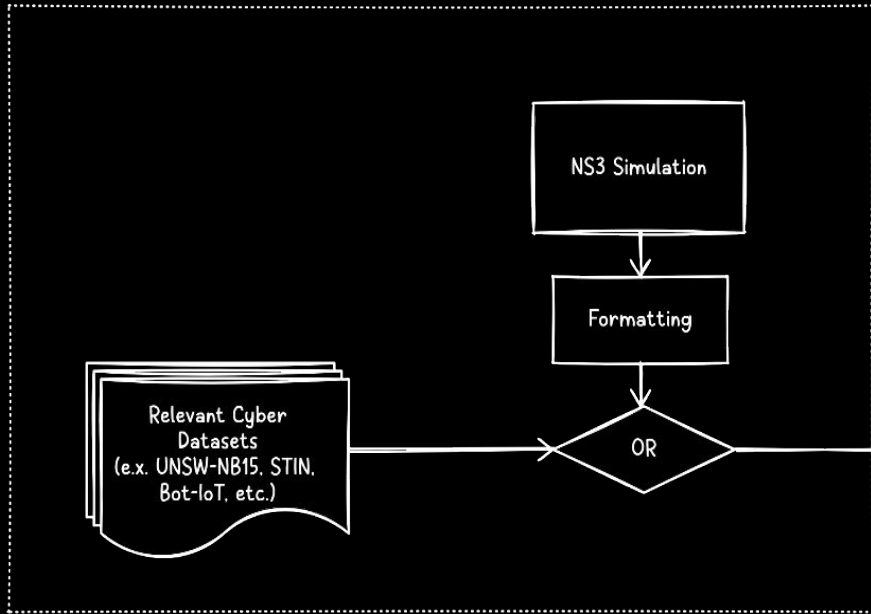
ns-3 Simulation

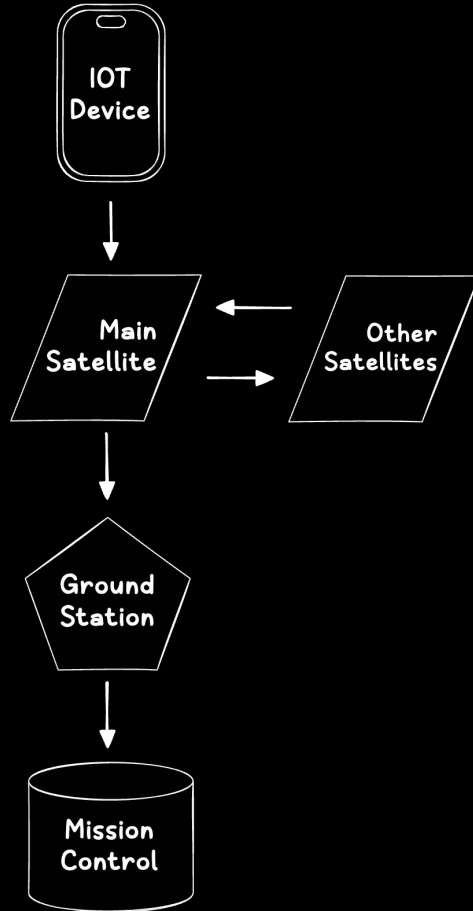
Why ns-3?

- Chosen for flexibility and low-level control
- Enables simulation of space-relevant network conditions
- Supports reproducible and tagged packet trace generation
- Supports adding modules: satellite module & satellite protocol rulesets (CCSDS Blue Books)

Simulation Input Flexibility

- Supports real or synthetic traffic inputs
- UNSW-NB15 used for early testing
- PCAP & CSV formats used for trace generation
- Formatting aligns data for post-simulation evaluation



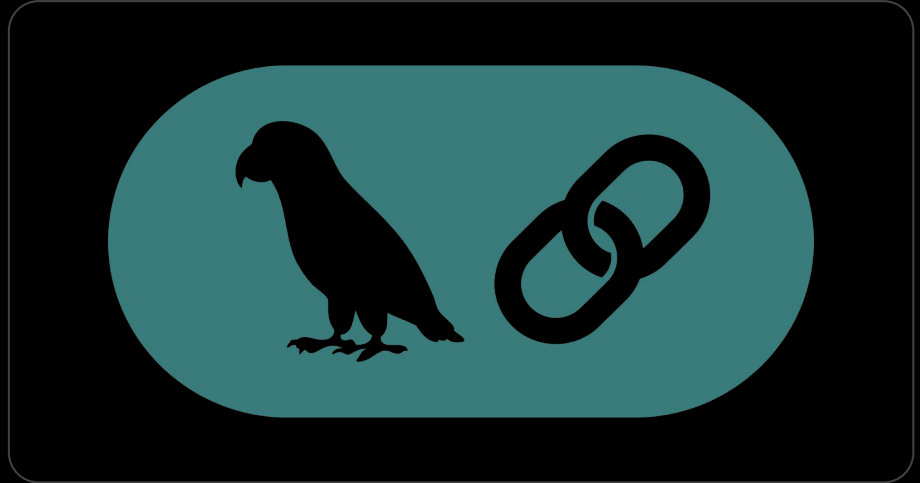
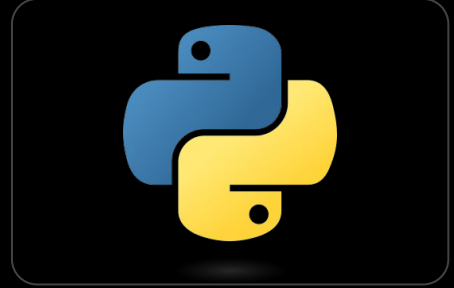
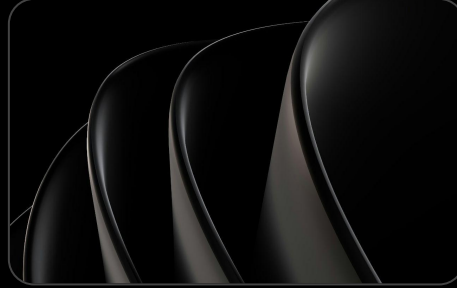


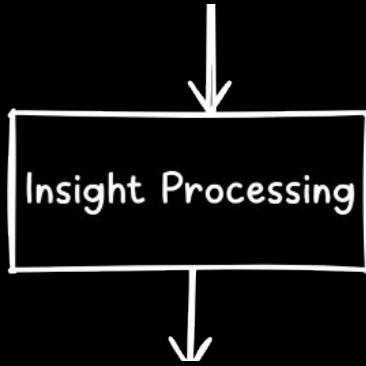
ns-3 System Overview

- Simulates data flow from IoT devices through space-ground infrastructure
- Would model realistic behavior: latency, relay, packet loss
- Core components: IoT-> Satellite-> Ground Station -> Mission Control
- Modular Setup allows for edge node expansion

AI Pipeline

The AI pipeline takes data, generates insights on it, and converts it into a natural language representation. It then takes some pretagged NL data, NL data to be evaluated on, and forms them into a prompt using a boilerplate. Finally, inference is run on this prompt by one or more agents, output is parsed, and a loss metric is run to evaluate performance.

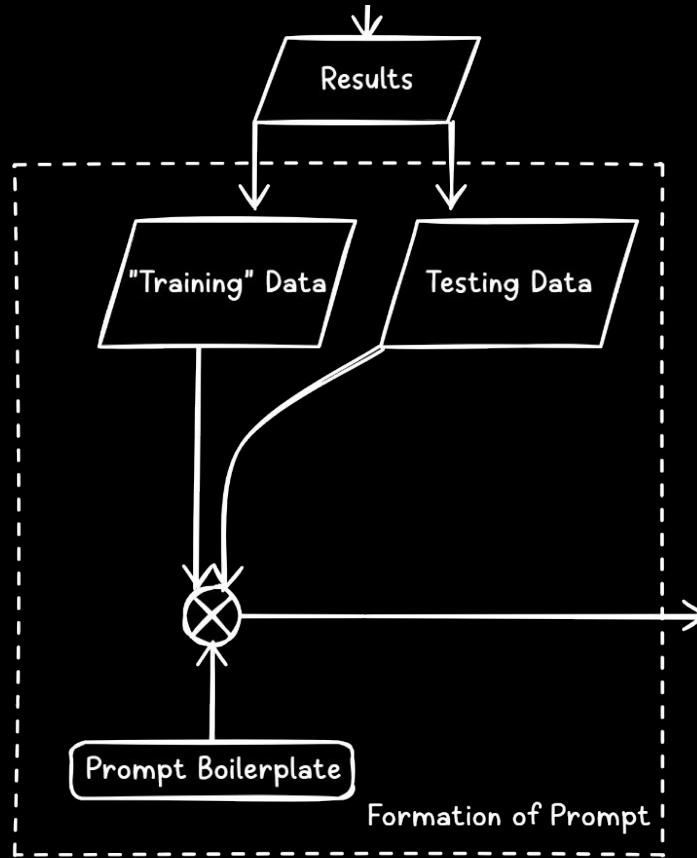




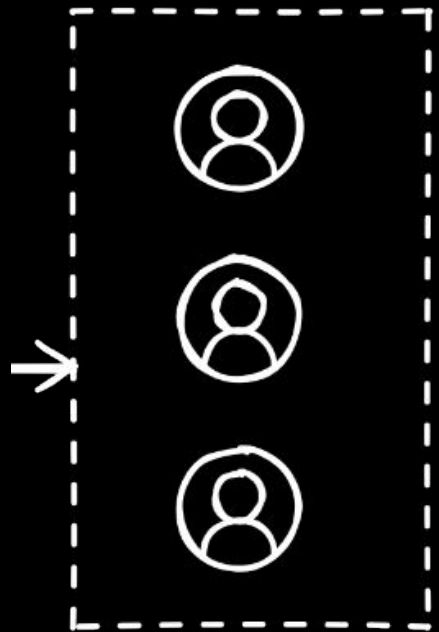
Insight Processing

- Takes raw packet data
- Natural Language Representation
- Flow-based sampling
- Generation of Natural Language Insights

Formation of Prompt



- Split input data into pre-tagged "train" and "test" data*
- Use prompt boilerplate
- Output combined prompt



Evaluate
Agent Collection

Evaluate Agent Collection

- Perform inference
- Evaluate performance based on metric
 - Correct Format
 - Correct identification of attack
 - Flagging False-Positive rate
 - Flagging False-Negative rate
 - Categorization of attack
 - Inference Time

Areas of Concern

X-Interaction

Cross-module protocol

Bridge

- Complex needs of project needs a suitable bridge
- Test early, often, and comprehensively

Output Format

Valid LLM output

AI

- Obtaining structured output may be difficult
 - Small model size = Lower quality output
- Optimization based fixes
 - Prompt Optimization
 - Fine-Tuning Optimization
- Representation based fixes
 - Smaller
 - Specialized

Data Validity

Nonsensical results

ns-3

- Identified as a strong possibility with original resimulation design by Dr. Daniels
- Cause/Effect violated
- Required reevaluation of initial design
- New design uses original application
- Still a possibility
- Care must be taken to ensure accurate function

Design Analysis

ns-3 Simulation

Current State / Future Exploration

- Satellite node design guided by CCSDS standards for realism
- IoT, ground stations, and edge units under active exploration
- Focus on simulating latency, signal degradation, and routing complexity
- Modular components allow for independent validation
- Challenges help inform architectural refinement

AI Pipeline

Current State / Future Exploration

- Base Pipeline implemented utilizing LangChain
- Takes a single prompt, and provides it to available models
- Allows for Inferencing on multiple models in parallel
- Set for rapid future development

Testing

Unit Testing

- ns-3: Validates MAC/PHY layer setup, packet flow, orbital scheduling.
- LLM: Ensures prompt generation, data splitting, output parsing logic works as expected.

Integration Testing

- Full-stack test from IoT simulation to LLM anomaly classification.
- Scenarios include normal traffic, spoofing, and signal interference patterns.

Interface Testing

- Tests seamless data conversion (PCAP/CSV → insights → prompts).
- Validates compatibility between ns-3 output and LLM input.

Tools & Automation

- Python-based test harness
- CI/CD pipeline for regression & consistency checks

System Testing

- Evaluates detection accuracy, latency, and false positive/negative rates under stress (e.g., high-latency, low-bandwidth).

Implementation

ns-3 Simulation Module

- Custom application in ns-3 simulates data flow from IoT devices → Satellites → Ground Station → Mission Control.
- Models realistic latency, signal degradation, and routing logic.
- Outputs PCAP + CSV data for analysis.
- Satellite module designed using CCSDS Blue Book standards.

Current Status

- Base simulation (no modules) and AI pipeline are implemented.
- Ongoing: Full satellite node integration, optimizing data flows, and enhancing real-time performance

LLM Evaluation Pipeline

- Built using **LangChain + Hugging Face** for local, parallel LLM evaluation.
- Converts simulation outputs into **natural language insights** and formats them into prompts.
- Supports multiple models running inference simultaneously.
- Includes prompt templating, insight generation, and output parsing.

Thank You



Summary

Questions?

Let's talk.

1



Network Simulator (ns-3)

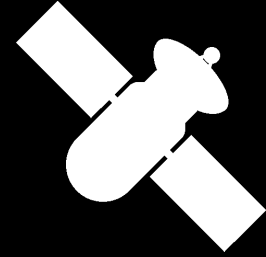
Simulating deep space
telecommunication network.

2



LangChain + Hugging Face

Platforms for building and maintaining
applications that utilize LLMs.



Real-time Anomaly
Detection in Space
Communications using
Large Language Models